

Interactive Digital Art and Design SE608

Project Proposal Year IV



Student Name:	Yullia Antonova	Student Number:	C00301432
Working Title:	Interactive Generative Digital Plant		
Description:	This project will create an interactive generative 3D digital plant that grows and develops over time. Different parts of the plant, such as stems, branches, leaves and flowers, will be generated and controlled procedurally rather than being created as one fixed model. The main technical challenge will be to develop a system that can generate and modify the plant structure using programmed rules. I will research procedural plant and tree generation methods, including the Space Colonization Algorithm and L-systems, and investigate which approach is most suitable for my project. I will use Autodesk Maya to create the initial 3D geometry and reusable assets, such as leaves, stems and flowers. These assets will then be controlled and positioned by code to create the final plant structure. JavaScript and Three.js will be used to develop the interactive 3D system, while WebGL will provide real-time rendering in the browser. User interaction through the mouse and keyboard will influence selected growth parameters and the behaviour of the plant. Animation will be used to visualise the growth and movement of the generated plant. The main focus of the project will therefore be procedural generation and user interaction, with animation supporting the visualisation of the growth process.		
Reasons for selecting project:	I am interested in digital art, animation and interactive experiences, and I am also inspired by natural forms and plants. This project allows me to combine these interests while developing my technical skills in 3D modelling, procedural generation, animation and creative coding. I am particularly interested in exploring how a digital plant can be made to appear alive through growth, movement and interaction. I want to investigate how geometry and animation can be controlled through rules and code, and how user input can influence the behaviour and appearance of the plant. This project will allow me to combine artistic creativity with technical problem-solving and develop my skills in creating an interactive digital experience. The project will also allow me to develop my technical skills in procedural generation, 3D modelling, JavaScript programming and real-time 3D interaction. I am particularly interested in investigating how programmed rules can be used to reproduce		

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	aspects of natural growth and transform them into an interactive digital experience.
Proposed research content:	I will research interactive and generative digital art, with a particular focus on procedural generation of natural forms and interactive 3D environments. I will investigate how programmed rules can be used to create structures that develop and change over time rather than relying on a fixed animation. I will research procedural methods for generating plants and tree-like structures. In particular, I will investigate the Space Colonization Algorithm as a possible method for generating branching structures. I will also research L-systems as an alternative rule-based approach to plant generation. The research will help me determine which method is most appropriate for creating the structure of my digital plant. I will investigate how 3D geometry can be created and prepared for procedural generation. Autodesk Maya will be used to model reusable components such as leaves, stems and flowers. I will explore how these components can be positioned, rotated and scaled by code rather than manually placing every element of the plant. I will research JavaScript and Three.js for creating the interactive 3D environment. Three.js will be used to create and control the 3D scene, while WebGL will provide real-time rendering in the browser. I will investigate how user input from the mouse and keyboard can be connected to parameters controlling plant growth, movement and appearance. I will also research animation techniques for visualising procedural growth. The animation will support the procedural system by showing changes in the plant over time rather than being the main focus of the project. The final research will therefore combine generative art, procedural plant generation, 3D modelling, JavaScript programming, real-time WebGL rendering, animation and user interaction. Proposed technical process 3D assets created in Maya ↓ Procedural growth algorithm ↓ JavaScript / Three.js ↓

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	Branches and plant elements generated dynamically ↓ User interaction changes growth parameters ↓ WebGL renders the plant in real time Create an interactive browser-based 3D plant that develops procedurally while the program is running. The initial geometric elements are created in Maya, and JavaScript/Three.js controls their procedural placement and growth. WebGL is used for 3D rendering. The user interacts with the system via the mouse and keyboard to influence the plant's behavior. The Space Colonization Algorithm serves as the primary research foundation, while L-systems are considered as an alternative approach. The main technical challenge is not to animate a pre-made tree, but to create an algorithmic system capable of dynamically generating and modifying the plant's structure.
External links (if applicable):	Magic Ivy – Interactive 3D Plant https://sites2.org.aalto.fi/medialabdemodays_old/projects/magic-ivy/ An interactive and generative 3D plant created using JavaScript and Three.js. The plant is generated with code and responds to user interaction. I will use this project as a reference for combining procedural plant growth with interactive 3D visuals. BOTanic Garden – A p5.js Generative Artwork https://www.behance.net/gallery/76353549/BOTanic-Garden-A-p5js-generative-artwork A generative plant simulation created with p5.js. The plants grow, reproduce and die according to programmed rules, creating a changing digital environment. I will research this project to understand how natural plant growth can be represented using code. All A Garden – Interactive Art https://nesrasportfolio.com/all_garden An interactive botanical artwork created with p5.js. User movement affects the growth of flowers and thorns. I will use this project as a reference for exploring how user interaction can change the behaviour and appearance of digital plants.

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	Space Colonization Algorithm Runions, A., Lane, B. & Prusinkiewicz, P. (2007). Modeling Trees with a Space Colonization Algorithm. DOI: 10.2312/NPH/NPH07/063-070 L-systems Prusinkiewicz, P. & Lindenmayer, A. — The Algorithmic Beauty of Plants
Hardware requirements:	Computer with keyboard and mouse.
Software requirements:	Autodesk Maya, JavaScript, Three.js, WebGL, HTML/CSS, p5.js, Adobe Photoshop and GitHub.
Other requirements:	Internet access for research and testing.

Signed: Yuliia Antonova

Date: 16.09.2026

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Approved/Not Approved:	
Reasons for not approving project:	
Conditions attached to approving project:	
Approved/Not Approved:	
Name of Supervisor:	

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Signed: _____	Date: _____